Analysis of Titanic Dataset

Introduction

The Titanic dataset is a historical record of passengers aboard the RMS Titanic, which sank in 1912. This analysis explores passenger demographics, survival rates, and correlations between features to uncover patterns and insights.

Data Cleaning

1. Missing values in `Age` were filled with the median.

2. Rows with missing `Embarked` values were dropped.

3. Missing values in `Cabin` were labeled as "Unknown."

4. A new feature `FamilySize` was created from `SibSp` and `Parch`.

Visualizations

1. Age Distribution (Histogram):

   - A bimodal distribution reveals peaks among children and young adults.

   - Highlights the diversity of passenger age groups.

2. Fare Across Passenger Classes (Boxplot):

   - Higher fares correspond to first-class passengers.

   - Outliers indicate exceptionally high fares paid by some passengers.

3. Age vs. Fare (Scatterplot):

   - No clear linear trend observed between `Age` and `Fare`.

   - Survivors often correlate with higher fares.

4. Pairplot of Numerical Features:

   - Highlights relationships between features such as `Fare`, `Age`, and `Pclass`.

   - Clusters suggest demographic trends.

5. Correlation Matrix (Heatmap):

   - Strong negative correlation between `Fare` and `Pclass`.

   - Moderate correlation between `SibSp` and `Parch`, representing family groups.

Observations

- Age distribution and survival rates suggest the importance of demographics in survival outcomes.

- Higher passenger classes are strongly correlated with survival.

- Family size provides additional context for understanding passenger group dynamics.

Conclusion

This analysis highlights key survival factors aboard the Titanic, including class, fare, and age. The insights can inform predictive models, aiding future exploration of historical datasets.